

Project Communication & Presentation Plan

1. Presentation Structure

The final demonstration is designed as a structured, 10-slide slide presentation targeting stakeholders, investors, and agricultural administrators.

Slide-by-Slide Outline:

- **Slide 1: Title Slide:** Project Name (Opti-Crop), Team Roles, and Vision.
- **Slide 2: The Core Problem:** Soil degradation, high fertilizer costs, and crop failure.
- **Slide 3: Our Proposed Solution:** Data-driven crop prediction app featuring immediate corrective soil guidance.
- **Slide 4: System Architecture:** Three-tier architecture (Presentation, Flask Backend, Relational SQLite DB).
- **Slide 5: Machine Learning Insights:** phosphorus outlier removal, Logistic Regression training (94% accuracy), and KMeans clustering elbow curve.
- **Slide 6: Live Web Application Demo:** Video or walk-through of crop recommendation flow.
- **Slide 7: Soil Suitability Calculator:** Demo of testing custom crops with z-score analytics.
- **Slide 8: User Login & Log History:** Dashboard view showing previous soil reports.
- **Slide 9: Scalability & Future Scope:** Real-time IoT sensor integrations, Weather API linkages, and mobile development.
- **Slide 10: Conclusion & Q&A:** Final thoughts, impact summary, and opening for questions.